Improving Classification Accuracy For Diabetic Retinopathy With Deep Learning

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Abstract. Diabetic retinopathy (DR) is a leading cause of vision impairment worldwide, emphasizing the critical need for early detection and diagnosis to prevent irreversible damage. Deep learning (DL) has emerged as a transformative approach in medical imaging, providing enhanced accuracy and efficiency in disease classification tasks, including DR detection. Despite the promising advancements, existing models for DR classification often face challenges such as limited generalization across diverse datasets and inadequate performance in real-world scenarios. This study proposes a novel approach utilizing SqueezeNet with Squeeze-and-Excitation (SE) blocks to improve DR classification. This architecture is designed to optimize computational efficiency while enhancing feature extraction capabilities, thus addressing the limitations of traditional models. Experiments demonstrate that the proposed model achieved a test accuracy of 90.9%, significantly outperforming previous benchmarks in DR image classification. The results underscore the potential of leveraging advanced DL techniques to enhance diagnostic accuracy, paving the way for more effective clinical applications in the early detection of diabetic retinopathy.

INTRODUCTION

Diabetic retinopathy (DR) is a retinal disorder affecting individuals with prolonged diabetes, resulting from the damage and weakening of retinal blood vessels due to elevated blood sugar levels.[1] As a result, microaneurysms, exudates, hemorrhages, and cotton-wool spots manifest.[2] Microaneurysms, exudates, retinal hemorrhages, and cotton-wool spots are progressive retinal signs of diabetic retinopathy, caused by fluid and blood leakage from weakened vessels, indicating escalating risk of vision loss.[3] Diabetic retinopathy, a leading cause of blindness, requires early diagnosis, which has driven the development of automated systems due to the increasing number of patients and the limited availability of skilled technicians.[4] Diabetic retinopathy (DR) is a progressive retinal disease caused by prolonged high blood glucose and blood pressure in diabetic individuals, making it a leading cause of preventable blindness worldwide.[5] Approximately one-third of diabetic patients develop DR, which damages the retinal microvasculature and leads to abnormalities like microaneurysms, haemorrhages, exudates, and capillary closures, ultimately causing vision loss.[6] The World Health Organization (WHO) projects diabetes to become the 7th leading cause of death, with the global diabetic population rising from 108 million in 1980 to 422 million by 2014, particularly affecting those below the poverty line, and India alone is expected to see an increase from 61.3 million diabetic patients in 2023 to 101.2 million by 2030.[7]

The disease progresses through five stages, from no retinopathy to proliferative DR, with increasing severity marked by lesions, haemorrhages, and neovascularization, necessitating prompt treatment to prevent further damage.[8] The severity of diabetic retinopathy is determined by the presence and extent of retinal abnormalities. Key diagnostic indicators include microaneurysms, haemorrhages, neovascularization, and venous beading. Microaneurysms are small blood clots, 100–120 μm in size, while haemorrhages result from blood vessel rupture.[9] Neovascularization refers to abnormal new blood vessel growth, and venous beading indicates the dilation of veins near occluded arterioles. Diabetic retinopathy is classified into Non-Proliferative Diabetic Retinopathy (NPDR) and Proliferative Diabetic Retinopathy (PDR), with NPDR further divided into mild, moderate, and severe stages based on disease progression.[10]

Early detection of diabetic retinopathy (DR) is paramount, as timely intervention can prevent the visual impairment and blindness associated with the disease.[11] Despite DR's progressive nature, its debilitating effects can be mitigated with prompt diagnosis and treatment. Regular digital fundus imaging plays a crucial role in identifying early signs of DR, allowing for immediate management before irreversible damage occurs. Clinical guidelines stress the importance of lifelong, routine screening for diabetic patients, with more frequent examinations recommended for those in advanced stages of the disease.[12] However, the rising global prevalence of DR and the strain on healthcare systems underscore the necessity of efficient early detection methods to alleviate the burden on clinicians and ensure effective patient care.[13]

Conventional methods for diagnosing diabetic retinopathy (DR) typically rely on manual inspection of retinal fundus images by trained ophthalmologists or employ image processing techniques that use handcrafted features to identify abnormalities.[14] These methods, though effective to a certain extent, are limited by their dependence on human expertise and the difficulty of accurately capturing the complex and subtle patterns associated with different stages of diabetic retinopathy.[15] Furthermore, the variability in image quality, patient diversity, and disease progression presents challenges for traditional approaches, often resulting in inconsistent and delayed diagnoses.[16]

Deep learning has emerged as a powerful technique for processing complex medical datasets, particularly in the realm of medical imaging.[17] By utilising convolutional neural networks, these models can automatically extract intricate features from diverse imaging modalities, enhancing the accuracy and efficiency of diagnoses across various medical conditions. In the context of diabetic retinopathy (DR), deep learning offers significant advantages by automating the detection of critical retinal abnormalities, such as microaneurysms, exudates, and haemorrhages.[18] This capability not only facilitates early diagnosis and precise monitoring of disease progression but also provides a scalable solution for mass screening, especially in underserved regions with limited access to specialised medical expertise. As deep learning technology continues to advance, its integration into clinical practice is poised to revolutionise the detection and management of DR, ultimately improving patient outcomes and accessibility to effective treatment.[19]

Shanthi and Sabeenian[8] conducted a study focused on diabetic retinopathy (DR) classification which employed a deep learning approach using a modified AlexNet architecture to categorize fundus images into different stages of the disease. The researchers applied a convolutional neural network (CNN) with Pooling, Rectified Linear Unit (ReLU) activation, and Softmax layers, which were essential for optimizing accuracy. This method was tested on the Messidor dataset, where 70% of the images were used for training and 30% for evaluation. The findings indicated high classification accuracies, with 96.6% for healthy images and similar results for DR stages, achieving 96.2% for stage 1, 95.6% for stage 2, and 96.6% for stage 3. These results demonstrate that CNNs are highly effective for DR classification, though the study suggests that increasing the dataset size or testing with larger datasets like Kaggle's could lead to further improvements.

Naz and Ahuja[20] introduced a web-based tool, RetinaCare, designed for the automated grading of diabetic retinopathy (DR) using a deep learning approach. The model employed a CNN architecture based on VGG-16, which was trained on the IDRiD and Messidor datasets to enhance DR detection accuracy. The method achieved significant performance metrics, with an accuracy of 96.89%, a specificity of 97.49%, and a sensitivity of 97.35%, indicating its high diagnostic capability. This study highlights the effectiveness of CNNs in addressing the need for accessible and efficient DR detection, particularly in resource-limited settings. Furthermore, the authors suggest future improvements could involve real-time case studies in remote areas to validate the tool's performance in practical scenarios.

Previous models for diabetic retinopathy (DR) classification often fall short in balancing accuracy and computational efficiency, especially when detecting early-stage signs such as microaneurysms and exudates. Many of these models rely on complex architectures or handcrafted features, which can be computationally intensive and struggle to generalize across diverse datasets, leading to inconsistent results. This paper presents a novel approach using SqueezeNet with a Squeeze-and-Excitation (SE) block, designed to address these challenges. The combination of SqueezeNet's lightweight architecture and the SE block's enhanced feature recalibration improves both accuracy and efficiency, making the model well-suited for scalable applications in clinical settings. The objective of this model is to provide a more resource-efficient yet highly accurate tool for the early detection and classification of DR, thereby improving diagnostic outcomes and accessibility to automated screening.

The paper is structured as follows: it begins with a Literature Review, followed by an outline of the Proposed Method. This is followed by the Experimental Results and Analysis, which includes subsections on Dataset Description, Implementation Details, and Results. Finally, the paper concludes with a comprehensive summary and a discussion of future work.

LITERATURE REVIEW

Deep learning (DL) is increasingly utilised across various medical fields due to its ability to automatically extract complex features from medical imaging data, improving tasks such as tumour detection, organ segmentation, and disease classification. Its success in analysing large-scale datasets makes it highly effective in areas like radiology, dermatology, and pathology. In the context of diabetic retinopathy (DR), DL has become crucial for early detection by identifying key retinal abnormalities, such as microaneurysms and exudates, from fundus images. This capability enables DL models to provide more accurate, scalable, and efficient solutions for DR screening and diagnosis, addressing the need for automated tools in healthcare.

Sagvekar et. al.,[21] developed the Hunter-Prey Ladybug Beetle Optimizer Deep Maxout Network (HPLBO_DMN) for diabetic retinopathy (DR) classification. The model involves preprocessing retinal images with a Kalman filter and Region of Interest (RoI) extraction, followed by lesion segmentation using K-Net, optimized by the Hunter-Prey Optimizer (HPO) and Ladybug Beetle Optimization (LBO). Feature extraction is enhanced with the Automatic Artery/Vein Classification Network (AVNet), and DR classification is performed using a Deep Maxout Network (DMN). Achieving 93.6% accuracy, 93.8% sensitivity, and 92.9% specificity, the model demonstrated strong performance but faces challenges with mild and moderate DR cases.

Sasikala et. el.,[22] proposed the Classification of Diabetic Retinopathy using Functional Linked Neural Network with Segmented Fundus Image Features (CDR-FLNN-SFI). This model processes input fundus images from the DIARETDB1 dataset, employing Federated Neural Collaborative Filtering for noise removal during preprocessing. Variational Density Peak Clustering is applied for segmentation, followed by classification using the Functional Linked Neural Network (FLNN). Achieving 99.87% accuracy, 98.34% precision, and a 97.41% F1-score, the model outperformed alternatives in both accuracy and computational efficiency. However, the study suggests further validation with larger datasets and proposes future integration of deep learning and meta-heuristic techniques for improved generalizability.

Teng et. al.,[23] MediDRNet, a dual-branch network model based on prototypical contrastive learning, was developed for diabetic retinopathy (DR) classification, addressing sample imbalance by focusing on lesion differences and similarities. The model's dual-branch structure transitions from feature learning, constructing a hierarchical system for lesion progression, to classification learning. MediDRNet incorporates the convolutional block attention module (CBAM) to enhance subtle retinal feature identification through spatial and channel attention. Experiments on Kaggle and UWF datasets demonstrated superior performance, particularly for minority DR categories. Future work aims to enhance the model's generalizability and efficiency in clinical settings for early DR diagnosis.

In the paper[24], Madarapu et. al. proposed the MuR-CAN model for diabetic retinopathy (DR) classification, addressing challenges like inter-class differences and small retinal lesions. The model incorporates a multi-dilation attention block (MDAB) that uses depth-wise convolutions with varying dilation rates to capture multi-scale spatial information, enhancing focus on key features. A support vector machine (SVM) is used for classification, leading to improved accuracy. MuR-CAN, combining CNNs like Xception and ResNet50V2, significantly outperforms state-of-the-art methods. Future work aims to tackle data imbalance using advanced augmentation techniques to enhance generalization and classification outcomes.

Abini and Priya's[25] study employed deep learning to classify diabetic retinopathy (DR) across normal, mild, moderate, severe, and proliferative stages using pre-trained CNNs, VGG-16, and MobileNet-V2. Aiming to overcome the limitations of manual feature extraction in previous research, the authors automated the classification process, achieving 90% and 92% accuracy in estimating DR stages. This approach effectively aids ophthalmologists in early DR detection, crucial for preventing severe visual impairments. The integration of advanced CNN architectures demonstrated significant advancements in DR diagnosis, enhancing diagnostic efficiency in clinical settings.

In a recent study[1], Madarapu et. al. developed a novel deep integrative approach for classifying diabetic retinopathy (DR) by addressing challenges like imbalanced datasets, small lesions, and inter-class similarity. Using residual blocks with a channel-spatial attention mechanism (CSAM), the model captures local and global information, while non-local blocks (NLB) further enhance important features and suppress irrelevant ones. This combination refines outputs and improves understanding of complex image relationships, leading to superior classification accuracy. The model also achieves efficiency with fewer trainable parameters and faster inference times than many state-of-the-art techniques. Experimental results show improved accuracy on comprehensive DR datasets.

In another study[26], Jagadesh et. al. proposed a novel two-stage approach for automated diabetic retinopathy (DR) classification, addressing challenges such as asymmetric datasets and small lesion detection. In the first stage, the improved contoured convolutional transformer (IC2T) segmented optical discs (OD) and blood vessels (BV), effectively capturing both local and global contexts. In the second stage, an Improved Coordination Attention Mechanism (ICAM) network identified key retinal biomarkers like microaneurysms and haemorrhages. The method was evaluated on EyePACS-1, Messidor-2, and DIARETDB0 datasets, achieving accuracy rates of 96%, 97%, and 98%, respectively, significantly outperforming state-of-the-art models.

METHODOLOGY

The increasing use of computer-aided technologies in the medical field has revolutionized the way healthcare is delivered. For instance, computer-aided diagnostics (CAD) systems leverage advanced algorithms to analyze medical images, assisting radiologists in detecting conditions such as tumors and fractures more accurately.[17] Additionally, electronic health records (EHR) streamline patient data management, enhancing the efficiency of healthcare delivery. Telemedicine platforms have also gained traction, allowing healthcare professionals to consult with patients remotely, thus expanding access to care, especially in underserved areas. Among the various applications of computer-aided technologies, diabetic retinopathy (DR) screening exemplifies a significant advancement. DR is a common complication of diabetes that can lead to severe vision loss if not detected early. Automated image analysis tools have been developed to assist in the diagnosis of DR by evaluating retinal images for signs of disease. These tools not only improve the accuracy of screenings but also facilitate timely interventions, ultimately enhancing patient outcomes. As technology continues to evolve, its integration into diabetes care is expected to expand, paving the way for more effective management strategies.

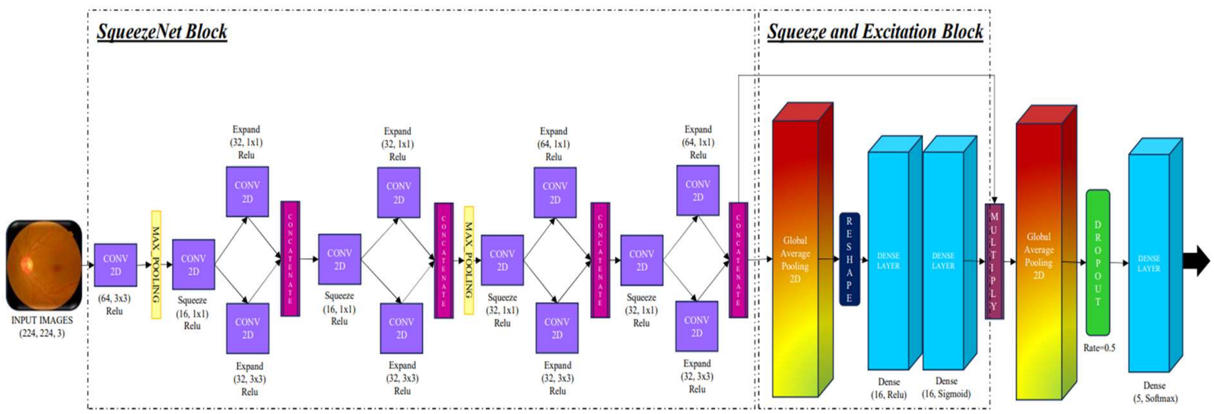


FIGURE 1. The architecture of SqueezeNet with Squeeze and Excitation Block

Deep learning (DL) and artificial intelligence (AI) have become pivotal in the advancement of medical diagnostics, particularly in the detection and classification of diseases such as diabetic retinopathy.[27] By utilizing convolutional neural networks (CNNs), DL algorithms can analyze vast amounts of retinal image data with remarkable accuracy. Transfer learning further enhances this capability by allowing pre-trained models to be fine-tuned on specific datasets. This approach is particularly beneficial in medical imaging, where acquiring large labeled datasets can be challenging. For example, a model trained on a diverse set of images from various sources can be adapted to recognize subtle features indicative of diabetic retinopathy in a smaller dataset, significantly reducing the time and resources required for model development. The key benefits of employing DL, AI, and transfer learning in this context are multifaceted. Firstly, these technologies enhance diagnostic accuracy by minimizing human error and increasing the consistency of image evaluations. Secondly, they improve the efficiency of screening processes, enabling healthcare providers to assess more patients in a shorter timeframe, thus facilitating early intervention. Lastly, by leveraging existing models through transfer learning, researchers can conserve computational resources and expedite the development of robust diagnostic tools, ultimately leading to better patient outcomes and more effective management of diabetic retinopathy. As the field continues to evolve, the integration of these technologies holds the promise of transforming diabetes care into a more precise and accessible domain.

The model employed in this research is based on SqueezeNet, a lightweight convolutional neural network architecture designed to achieve high accuracy while significantly reducing the number of parameters. By using a fire module, SqueezeNet compresses the number of input channels before applying filters, enabling efficient computation without sacrificing performance. This design is particularly advantageous in settings where computational resources are limited, such as mobile devices or edge computing applications. To enhance the model's performance further, a Squeeze-and-Excitation (SE) block is integrated into the SqueezeNet architecture. The SE block introduces a mechanism for channel-wise attention, allowing the network to focus on the most informative features while suppressing less relevant ones. This is achieved by first squeezing the feature maps to produce a channel descriptor, which is then passed through a learnable activation function. The resultant weights are multiplied back to the original feature maps, effectively enhancing the network's ability to capture important patterns in the data. The model's architecture is depicted in FIGURE 1.

The combination of SqueezeNet and the SE block offers several key benefits. Firstly, the model maintains a compact size, making it suitable for deployment in environments with limited resources while still delivering competitive accuracy. Secondly, the incorporation of the SE block improves feature representation by emphasizing the most critical channels, resulting in more robust classification performance. As a result, this model is well-suited for tasks such as diabetic retinopathy detection, where the ability to discern subtle variations in retinal images is crucial. By leveraging the strengths of SqueezeNet and the SE block, this approach provides an effective solution for the efficient classification of medical images, ultimately contributing to better patient care and outcomes.

The SqueezeNet model processes input images through a series of well-defined layers to evaluate and classify them efficiently. Initially, the model accepts a 224×224-pixel RGB image, which undergoes a 3×3 convolutional layer with ReLU activation to extract low-level features. This is followed by a max pooling layer that reduces spatial dimensions while retaining critical information. The core of the architecture consists of multiple fire modules, where each module applies a squeeze layer (a 1×1convolution) to compress the feature maps, followed by an expand layer that uses both 1×1 and 3×3 convolutions to capture diverse features. After processing through these modules, a Squeeze-and-Excitation block enhances feature representation by applying global average pooling to compute channel-wise attention, allowing the model to focus on the most informative features. Finally, a global average pooling layer reduces the dimensionality of the feature maps to a single vector, followed by dropout for regularization, and concludes with a softmax output layer that classifies the input image into one of five target classes. This layered approach enables the model to efficiently evaluate and classify medical images, making it particularly suitable for applications such as diabetic retinopathy detection.

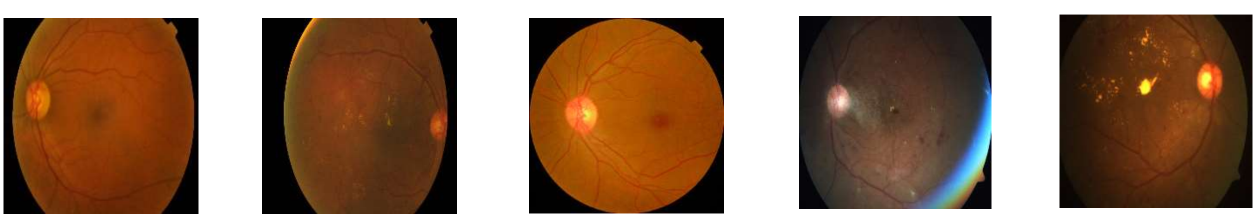
The SqueezeNet model with Squeeze-and-Excitation blocks stands out due to its unique architecture that effectively balances computational efficiency and high classification accuracy. Its innovative use of fire modules allows for a significant reduction in the number of parameters while maintaining robust feature extraction capabilities. Additionally, the integration of the Squeeze-and-Excitation block enhances the model's performance by enabling it to focus on the most relevant features, thereby improving the representational power of the learned features. This combination not only leads to faster inference times but also results in improved accuracy, making it particularly effective for complex tasks such as diabetic retinopathy detection. These advantages underscore the decision to adopt this model, as it aligns with the need for a lightweight yet powerful solution in medical image classification, ultimately contributing to better diagnostic outcomes and patient care.

Dataset Description

The dataset[28] used in this study consists of 3662 retina scan images, resized to 224x224 pixels for compatibility with pre-trained deep learning models; some sample images are displayed class wise in

(a) (b) (c) (d) (e)

FIGURE 2. Sourced from the APTOS 2019 Blindness Detection challenge, the dataset is well-organized, with images classified into five distinct categories based on the severity of diabetic retinopathy. These categories include No_DR (1805 images, representing healthy eyes), Mild (370 images, indicating early-stage retinopathy), Moderate (999 images, showing intermediate severity), Severe (193 images, highlighting advanced retinopathy), and Proliferate_DR (295 images, reflecting the most severe stage). The balanced structure and clear categorization of this dataset provide a robust foundation for training models to detect varying stages of diabetic retinopathy, making it an



(a) (b) (c) (d) (e)

FIGURE 2. Sample images from the Dataset: (a) is the Mild class, (b) is Moderate class, (c) is No_DR class, (d) is Proliferative_DR class and (e) is Severe class

ideal choice for this research.

Data Preprocessing

In this research, the dataset comprising retina scan images for diabetic retinopathy detection was resized to 224×224 pixels to ensure compatibility with pre-trained deep learning models, facilitating smoother integration and faster processing. Upon analyzing the dataset, it was evident that there was a significant class imbalance, with certain classes like No_DR being overrepresented compared to others such as Severe and Proliferate_DR. To address this, class balancing techniques were employed to ensure that the model received an equal representation of all severity levels during training. TABLE 1 shows the distribution of images before and after augmentation. This step is crucial as it prevents the model from being biased towards the majority class, leading to more accurate and generalized predictions. Once the class imbalance was rectified, the dataset was split into training, validation, and test sets in an 80:10:10 ratio, ensuring a proper distribution for model training, validation, and evaluation. TABLE 2 shows the number of images in train, test, and validation datasets and

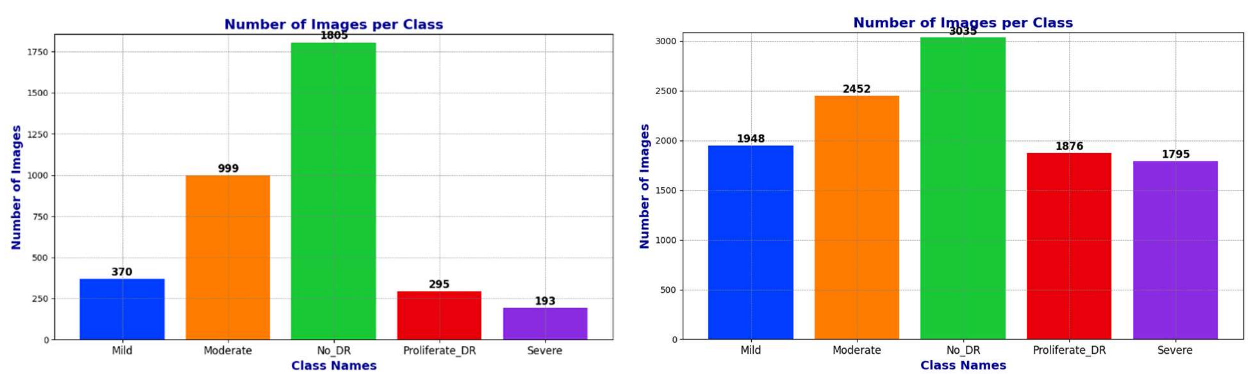
(a) (b)

FIGURE 3 shows the visualization of the same in the form of bar chart. Data augmentation was then applied to the training set using a variety of transformations, including rotation (up to 20 degrees), width and height shifts (up to 20%), shear, zoom, horizontal flips, and nearest neighbor filling. These augmentations helped introduce variability in the dataset, improving the model's ability to generalize to unseen data by simulating real-world variations. This not only reduced overfitting but also enhanced the robustness of the model, allowing it to perform better on diverse retinal images.

TABLE 1. Distribution of Images in Dataset

Distribution Before Augmentation		Distribution After Augmentation	
Classes	**Number of Images**	**Classes**	**Number of Images**
Mild	370	Mild	1948
Moderate	999	Moderate	2452
No_DR	1805	No_DR	3035
Proliferative_DR	295	Proliferative_DR	1876
Severe	193	Severe	1795

TABLE 2. Distribution of Images for Training, testing and Validation



(a) (b)

FIGURE 3. Bar Chart visualization of image dataset Before and After Augmentation

Dataset	Number of Images
Train	8883
Test	1110
Validation	1113

Implementation Details

The system utilized for this project is specifically designed to handle computationally intensive workloads, equipped with an NVIDIA RTX 3050 graphics card featuring 4GB of dedicated memory for parallel processing and image computation. This, coupled with a 12th Gen Intel Core i7 processor running at 2.30 GHz and 16GB of RAM, ensures smooth multitasking and efficient management of large datasets without performance bottlenecks. Built on a 64-bit architecture with Windows 11, the system provides a robust platform for deep learning tasks, allowing for seamless execution of machine learning models. In TABLE 3 the specifications of the system are listed. This high-performance hardware, combined with ample memory, is well-suited to the resource demands of deep learning tasks such as image classification.

TABLE 3. System Specifications

Name	Parameter
GPU	4GB
Graphics Card	RTX 3050
Processor	12th Gen Intel Core i7
Ram	16GB
Operating System	Windows 11
Architecture	64-bit

The programming environment consists of VS Code for development and evaluation, with TensorFlow and Python as the core tools. The model is trained on images resized to 224×224 pixels, optimizing memory use while maintaining sufficient detail for accurate classification. A batch size of 8 is used, ensuring efficient GPU utilization and faster training. The Adam optimizer, known for its adaptive learning rates, is applied alongside the categorical cross-entropy loss function, well-suited for multiclass classification tasks. The model is trained for 50 epochs, allowing it to learn effectively without overfitting, across 5 distinct classes. These choices of hyperparameters were made to ensure a balanced trade-off between computational efficiency and model performance, leading to more accurate and generalizable results.TABLE 4 displays the hyperparameters of the model.

TABLE 4. Description of Hyperparameters

Hyperparameter	Value
Image Size	224×224
Batch Size	8
Optimizer	Adam
Loss Function	Categorical Cross entropy
Epochs	50
Classes	5

In conclusion, the combination of a carefully curated dataset, effective preprocessing techniques, and an optimized implementation has significantly contributed to the performance of the SqueezeNet model enhanced with Squeeze-and-Excitation blocks. The preprocessing steps, including image resizing and class balancing, ensured that the model could handle the data efficiently while maintaining accuracy across all categories. The application of data augmentation further strengthened the model's ability to generalize well to unseen data by simulating various real-world conditions. Additionally, the implementation on a high-performance system allowed for the seamless execution of computationally demanding tasks, enabling the model to achieve superior results in diabetic retinopathy detection. These thoughtful considerations have made the model not only lightweight but also highly effective in delivering accurate and reliable classifications.

EXPERIMATAL RESULT AND ANALYSIS

The SqueezeNet model enhanced with a Squeeze-and-Excitation (SE) block demonstrated strong performance in the classification of diabetic retinopathy images. The model achieved a training accuracy of 92.72% at the 50th epoch, with a testing accuracy of 90.9% and a validation accuracy of 91.37%. These results underscore the effectiveness of the SE block, which adaptively recalibrates channel-wise feature responses, leading to improved model precision. The relatively small size of the SqueezeNet architecture, combined with the SE block's ability to focus on relevant features, provided a good balance between model complexity and performance. This is particularly advantageous for medical image classification tasks where overfitting is a concern, yet high accuracy is critical. The trends in training and validation loss shown in

(a) (b)

FIGURE 6.(a) and the corresponding accuracy graphs in

(a) (b)

FIGURE 6.(b) further demonstrate the model's generalizability and stability across both training and validation phases. The

(a) (b)

FIGURE 6 image displays the training and validation loss and accuracy over time. Both training and validation loss decreased steadily, indicating the model's effective learning process. Accuracy consistently improved, with training and validation accuracy reaching around 90%. This demonstrates that the model is highly accurate and performs well on unseen validation data, showcasing its reliability in real-world applications. **Error! Reference source not found.FIGURE 4** shows the ROC curves for several classes, with the AUC values all ranging from 0.97 to 1.00. These curves reflect excellent classification performance, as the model accurately differentiates between classes. The AUC score of 1.00 for No_DR class demonstrates perfect classification for that specific category. Overall, the model effectively distinguishes between all classes, providing reliable and confident predictions. FIGURE 4 represents a confusion matrix, illustrating the model's performance across the five classes. A high number of correct predictions are observed along the diagonal, especially for No_DR, Proliferative_DR, and Severe, indicating that the model is highly accurate in classifying these categories. This result emphasizes the model's ability to focus on relevant features in each class, aided by the Squeeze-and-Excitation (SE) block, which enhances the feature recalibration process and improves classification performance. FIGURE 8 displays the probability distribution of predicted probabilities for each class. The model shows a clear separation between the probabilities assigned to each class, particularly for No_DR and Severe, where the peaks are sharper. This reflects the model's confidence in its predictions, with the SE block playing a significant role by ensuring the model focuses on the most relevant features of the retinal images. The FIGURE 7 highlights the class-wise precision, recall, and F1-score metrics, demonstrating strong performance across all five classes. No_DR and Severe achieved particularly high scores, confirming the model's effectiveness in accurately identifying these classes. The precision and recall scores for other classes are also noteworthy, showcasing the model's robustness in detecting diabetic retinopathy across varying severity levels. The

(a) (b)

FIGURE 6. Performance metrics per epoch illustrate training and validation accuracy and loss

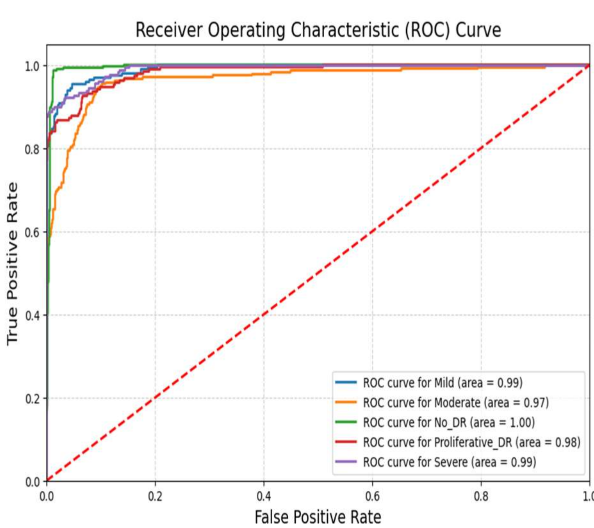


FIGURE 5. ROC Curve

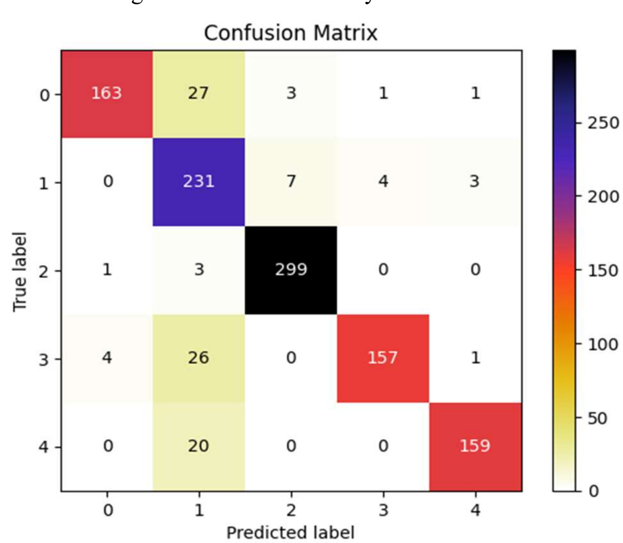


FIGURE 4. Confusion Matrix

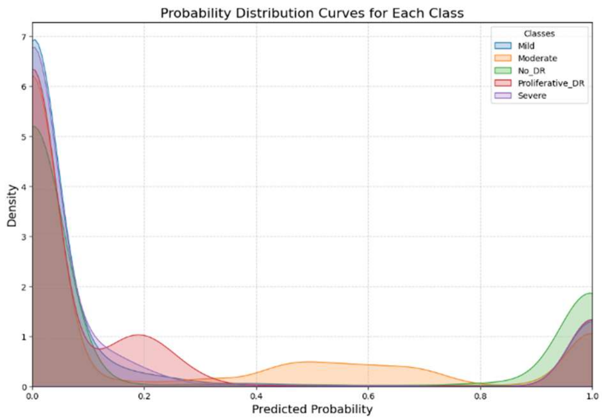


FIGURE 8. Probability Distribution Curve

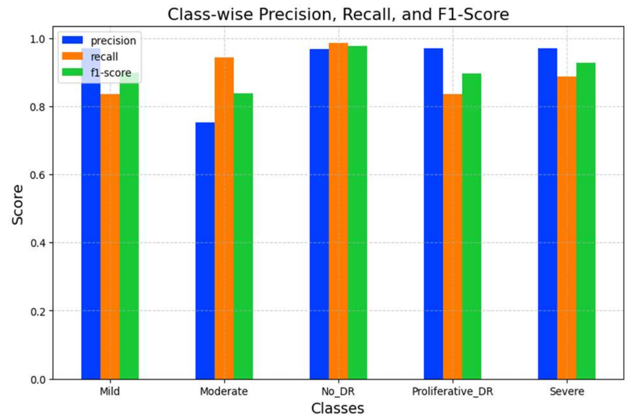
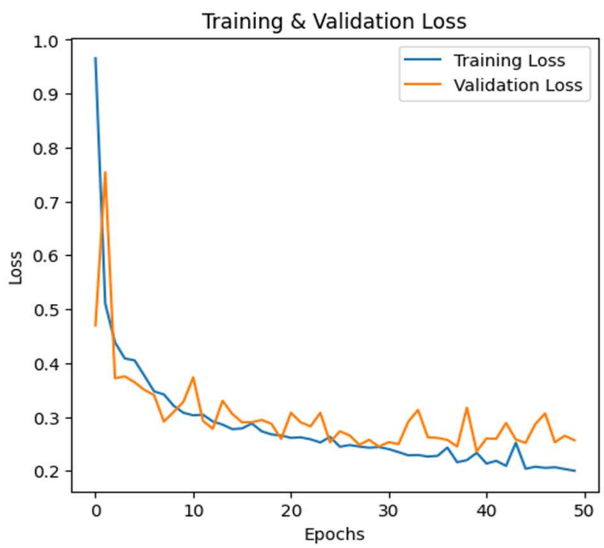
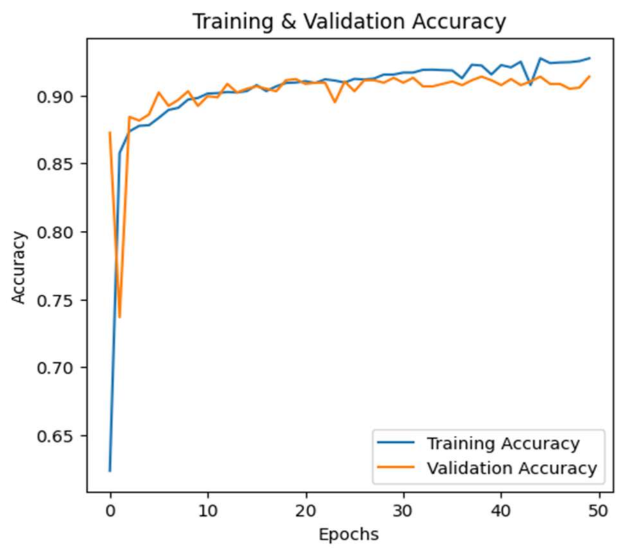


FIGURE 7. Class-wise Precision, Recall and F1-Score

SE block's contribution is evident in improving the model's ability to handle subtle differences between classes, ultimately enhancing the overall classification quality.

The TABLE 5 compares the performance of various models on diabetic retinopathy detection in terms of accuracy, precision, recall, and F1-score. Wong et al. achieved 82.10% across all metrics, while Guanghui et al. demonstrated a higher accuracy of 83.40%, though with notably lower precision, recall, and F1-score. Bodapati et al.'s two models show improvements, with the second model achieving 86.22% accuracy, though with slightly lower recall and F1-score. The proposed model outperforms all prior models with a notable 90.90% accuracy, 93.00% precision, and balanced recall and F1-score, highlighting its effectiveness in achieving more accurate and reliable classifications.





The model demonstrated strong performance across all evaluation metrics, consistently achieving high accuracy and minimal loss in both training and validation phases. The ROC and probability density functions reinforced the model's ability to make confident predictions across different classes, showing a clear distinction in probability distributions. The confusion matrix and class-wise precision-recall metrics further highlighted the model's precision in identifying diabetic retinopathy stages, particularly for the most critical classes. The use of the Squeeze-and-Excitation (SE) block proved highly beneficial, enhancing feature extraction and ensuring that the model focused on the most relevant areas of the retinal images, ultimately leading to robust classification performance. These results align with the study's objective of efficiently classifying diabetic retinopathy using a lightweight architecture, demonstrating both accuracy and computational effectiveness.

TABLE 5. Comparison of Proposed Model with state-of -the-art models

Model	Accuracy	Precision	Recall	F1-Score
Wong et. al.[29]	82.10	82.10	82.10	82.10

Guanghui et. al.[30]	83.40	69.66	67.70	67.02
Bodapati et. al.[31]	84.31	84.00	84.00	84.00
Bodapati et. al.[32]	86.22	80.61	72.23	75.41
Proposed Model	90.90	93.00	90.00	91.00

CONCLUSION

In this research, we developed a SqueezeNet model with a Squeeze-and-Excitation (SE) block for the classification of diabetic retinopathy images from the APTOS dataset, aiming to enhance diagnostic accuracy and facilitate timely intervention in patients with diabetic retinopathy. The proposed model architecture efficiently balances low computational cost with high performance, leveraging the SE block to emphasize important features while suppressing less informative ones, thereby improving the model's classification capability.

The SqueezeNet with SE block demonstrated impressive efficiency, achieving a test accuracy of 90.9% across five classes: No DR, Mild, Moderate, Severe, and Proliferative DR. This level of performance indicates the model's robustness and its potential for deployment in clinical settings, where quick and reliable diagnoses are critical. The lightweight nature of SqueezeNet makes it particularly suitable for deployment on devices with limited computational resources, thus broadening access to advanced diagnostic tools. The successful application of this model highlights its benefits not only in providing accurate classifications but also in reducing the burden on healthcare professionals by automating the screening process. Moreover, this research contributes to the growing body of literature on utilizing deep learning techniques in medical image analysis, emphasizing the need for continued exploration of novel architectures and methods.

Future research directions may include refining the model through data augmentation techniques and exploring transfer learning approaches to further enhance performance on smaller datasets. Additionally, integrating the model with real-time imaging systems could revolutionize diabetic retinopathy screening processes, significantly impacting patient outcomes and healthcare efficiency. The findings of this study underscore the model's potential to transform diabetic retinopathy diagnosis and management, ultimately improving the quality of care for patients at risk.

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